# The curvature of the Fisher–Rao beta manifold decreases in both parameters

Defect calculus for Hessian metrics of additive differences
and a proof of the Le Brigant–Preston–Puechmorel conjecture

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**Abstract**

Let $K(x,y)$ be the sectional curvature of the Fisher–Rao metric on the manifold of beta distributions $\mathrm{Beta}(x,y)$, $x,y>0$. Le Brigant, Preston and Puechmorel showed that $K<0$, and they conjectured that $K>-\frac{1}{2}$; in the form recalled by Qi, the conjecture also asserts that $K$ is decreasing in each parameter. Previously only the diagonal $x=y$ had been settled. We prove the full conjecture: $\partial_x K<0$ and $\partial_y K<0$ on $(0,\infty)^2$, hence $-\frac{1}{2}<K<0$, and both bounds are sharp.

The proof rests on a *defect calculus* for Hessian metrics of additive differences $F(x)+F(y)-F(x+y)$. With $h=1/F''$ and $G=-F''/F'''$ the curvature factorizes as

$$K(x,y)=-\frac{h'(x)\,h'(y)\,h'(x+y)\,A(x,y)}{4\,B(x,y)^2},$$

where $A=G(x+y)-G(x)-G(y)$ and $B=h(x+y)-h(x)-h(y)$ are subadditivity defects. This identity gives four things. It gives a one-line proof of negativity. It gives exact edge laws $K(x,0^+)=-\frac{1}{2}\big(G'(x)-\frac{1}{2}\big)$ and $K(x,\infty)=-h'(x)\big(x-G(x)\big)/\big(4(x-h(x))^2\big)$. It gives a regular averaged formula for $\partial_x\log(-K)$ along the edges. And it gives a *defect balance* identity, $2B-A=\frac{1}{2}+O(\text{argument}^{-3})$, which removes the second-order cancellation at infinity.

For $F=\log\Gamma$, seven one-variable shape lemmas reduce the conjecture to finitely many inequalities in one variable plus one compact two-variable region. There, monotone corner bounds turn positivity on boxes into point evaluations. Every inequality is certified in ball arithmetic (Arb), with Alzer's enveloping series for large arguments; the complete verification runs in about two minutes.

As by-products we obtain the explicit squeeze $\max\{K(x,\infty),K(y,\infty)\}<K(x,y)<\min\{K(x,0^+),K(y,0^+)\}$, strict decrease along rays, and new proofs of the diagonal results of Qi and Qi–Agarwal.


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# 1 Introduction

## 1.1 The conjecture

The beta distributions $\mathrm{Beta}(x, y)$, with density $t^{x-1}(1-t)^{y-1}/B(x, y)$ on $(0, 1)$, form a two-parameter exponential family. Their Fisher information is the Hessian of the log-partition function

$$\varphi(x, y) = \log B(x, y) = \log\Gamma(x) + \log\Gamma(y) - \log\Gamma(x + y),$$

that is, with $\psi_k = \psi^{(k)}$ the polygamma functions and $s = x + y$,

$$g = \begin{pmatrix} \psi_1(x) - \psi_1(s) & -\psi_1(s) \\ -\psi_1(s) & \psi_1(y) - \psi_1(s) \end{pmatrix}. \tag{1}$$

The Fisher–Rao geometry of beta and Dirichlet distributions was studied by Le Brigant and Puechmorel [4] and by Le Brigant, Preston and Puechmorel [5]. They proved that the Dirichlet manifolds are Hadamard manifolds; in particular the sectional curvature $K(x, y)$ of the beta manifold is negative. Explicitly,

$$K(x, y) = \frac{\psi_2(s)\big[\psi_1(x)\psi_2(y) + \psi_1(y)\psi_2(x)\big] - \psi_1(s)\psi_2(x)\psi_2(y)}{4\big[\psi_1(x)\psi_1(y) - \psi_1(s)(\psi_1(x) + \psi_1(y))\big]^2}. \tag{2}$$

From the limiting values and from numerical evidence they conjectured a lower bound [5]. The conjecture has been recalled in the following form in [7, 8].

**Conjecture 1.1** (Le Brigant–Preston–Puechmorel)**.** *The sectional curvature of the Fisher–Rao beta manifold satisfies $K(x, y) > -\frac{1}{2}$ for all $x, y > 0$, and $K(x, y)$ is decreasing in $x$ and in $y$.*

The progress so far concerns the diagonal. The sharp bounds $-\frac{1}{2} < K(x,x) < 0$ are established in [6, 7] through complete monotonicity of functions built from polygamma functions. In September 2026 Qi [8] proved that $x \mapsto K(x,x)$ is strictly decreasing. The two-variable statement remained open: it is recalled as a conjecture in [8], and the survey of open problems [10, §R2] lists it in October 2026 among the problems not attempted. The methods used on the diagonal (Laplace transforms, Bernstein's theorem) are one-dimensional, while $K$ depends on two variables through three polygamma arguments $x$, $y$, $x+y$.

### 1.2 Results

We write $h = 1/\psi_1$ and $G = -\psi_1/\psi_2$ (so $G = h/h'$), and

$$g(z) = z - G(z), \qquad e(z) = z - h(z) \qquad (z > 0).$$

Both $g$ and $e$ are increasing, concave, and tend to 0 at $0^+$ and to $\frac{1}{2}$ at $\infty$ (Corollary 3.4).

**Theorem A.** *For all $x, y > 0$,*

$$\frac{\partial K}{\partial x}(x,y) < 0 \qquad \text{and} \qquad \frac{\partial K}{\partial y}(x,y) < 0.$$

*Consequently $-\frac{1}{2} < K(x,y) < 0$ on $(0,\infty)^2$. Moreover $K(x,y) \to -\frac{1}{2}$ as $\min(x,y) \to \infty$ and $K(x,y) \to 0$ as $x+y \to 0$, so both bounds are sharp. Conjecture 1.1 is true.*

**Theorem B** (edge laws and squeeze)**.** *For every $x > 0$ the limits*

$$K_0(x) := \lim_{y\to 0^+} K(x,y) = -\frac{G'(x) - \frac{1}{2}}{2}, \qquad K_\infty(x) := \lim_{y\to\infty} K(x,y) = -\frac{h'(x)\,g(x)}{4\,e(x)^2}$$

*exist. $K_0$ is a strictly decreasing bijection of $(0,\infty)$ onto $(-\frac{1}{4}, 0)$, and $K_\infty$ is a strictly decreasing bijection onto $(-\frac{1}{2}, -\frac{1}{4})$. For all $x, y > 0$,*

$$\max\{K_\infty(x), K_\infty(y)\} < K(x,y) < \min\{K_0(x), K_0(y)\}.$$

**Theorem C** (defect calculus)**.** *Let $F \in C^4(0,\infty)$ with $F'' > 0 > F'''$, put $h = 1/F''$, $G = -F''/F'''$, and let $K_F$ be the Gaussian curvature of the Hessian metric of $\Phi(x,y) = F(x) + F(y) - F(x+y)$ wherever the metric is positive definite. Then*

$$K_F(x,y) = -\frac{h'(x)h'(y)h'(x+y)\,A(x,y)}{4B(x,y)^2},$$

*with $A, B$ as in the abstract. The metric is positive definite exactly where $B > 0$. Where $A > 0$, the logarithmic derivative $q = \partial_x \log(-K_F)$ is given by the raw formula (5), the balance formula (6) and, under a mild condition at 0, by the averaged formula (7).*

**Corollary D.** *The diagonal curvature $x \mapsto K(x,x)$ decreases strictly from 0 to $-\frac{1}{2}$ ([8], [6, 7]). For every $(x,y)$ the function $\lambda \mapsto K(\lambda x, \lambda y)$ is strictly decreasing on $(0,\infty)$. For $-\frac{1}{2} < c < 0$ each nonempty level set $\{K = c\}$ is the graph of a strictly decreasing smooth function.*

### 1.3 A new problem, the method, and the application

The proof follows a three-step scheme. First, a new problem forces the right structures to appear. Second, a method is developed to solve it. Third, the method is applied to Conjecture 1.1.

**Problem 1.2** (edge-squeeze problem)**.** Let $F$ be a strictly convex function on $(0,\infty)$ with $F''' < 0$, and let $K_F$ be the curvature of the Hessian metric of $F(x) + F(y) - F(x+y)$. Determine the edge laws $x \mapsto K_F(x, 0^+) = \lim_{y\to 0^+} K_F(x,y)$ and $x \mapsto K_F(x,\infty) = \lim_{y\to\infty} K_F(x,y)$. Decide whether $K_F(x,y)$ always lies strictly between its two edge values, and whether it is monotone between them.

For $F = \log\Gamma$, Theorem B answers Problem 1.2 completely. Since $K_\infty(x) \to -\frac{1}{2}$ as $x \to \infty$ and $K_\infty > -\frac{1}{2}$, the squeeze alone already gives the lower bound in Conjecture 1.1. The method has four ingredients.

(i) *Defect representation* (Theorem 2.2). The curvature is a quotient of two subadditivity defects, of $G$ and of $h$, weighted by $h'$ at the three arguments. This explains the sign of $K$, gives the edge laws, and reduces all two-variable information to two *one*-variable functions $G$ and $h$.

(ii) *Averaged representation* (Proposition 2.6). Writing $A$ and $B$ as averages of $G''$ and $h''$ over the segment family $\alpha x + \beta y$ gives a formula for $\partial_x \log(-K)$ in which the $1/x$ singularities along the edges have cancelled.

(iii) *Defect balance* (Proposition 2.4). At infinity $A \approx B \approx \frac{1}{2}$, but the first-order variations of $A$ and $B$ are locked in the ratio $2:1$: $2B - A = \frac{1}{2} + O(z^{-3})$. Rewriting $\partial_x \log(-K)$ through $\delta' = G' - 2h' + 1 = O(z^{-4})$ removes the second-order cancellation.

(iv) *Shape-constrained certification* (Sections 3–4). Seven one-variable shape lemmas (convexity of $h$ and $G$, unimodality of $G''$, signs of $h'''$ and of $2h'' - G''$, and so on) make every building block of $\partial_x \log(-K)$ monotone or unimodal in each variable. Positivity on a box then follows from finitely many point values at its corners. Outside a compact region the same lemmas reduce everything to inequalities in one variable.

### 1.4 Computer assistance

The shape lemmas and the remaining finitely many inequalities are verified with interval (ball) arithmetic. We use Arb [3] through python-flint, with exact rational symbolic preprocessing in SymPy. Three exact representations of each polygamma expression are used:

- an algebraically regularized form near 0, built from $\psi_k(1+z)$;
- the plain form on $[1, 3]$;
- Alzer's two-sided enveloping inequalities [1] on $[3, \infty)$, for which we include a short proof (Lemma 3.2).

No floating-point computation enters a proof. The scripts, the logs and a single command reproducing all certificates in about two minutes are described in Appendix A.

### 1.5 Organisation

Section 2 develops the defect calculus for general $F$ (Theorem C). Section 3 contains the polygamma toolkit and the shape lemmas for $F = \log\Gamma$. Section 4 proves that $\partial_x \log(-K) > 0$ by splitting $(0, \infty)^2$ into six regions. Section 5 proves Theorems A and B and Corollary D. Section 6 discusses asymptotics, the scope of the method and open questions.

## 2 Defect calculus for Hessian metrics of additive differences

### 2.1 Curvature of a two-dimensional Hessian metric

**Lemma 2.1.** *Let $\Phi \in C^4(U)$, $U \subset \mathbb{R}^2$ open, with positive definite Hessian $g = \nabla^2\Phi$, and write $\Phi_{ij}$, $\Phi_{ijk}$ for partial derivatives and $D = \det g$. The Gaussian curvature of $g$ is*

$$K = \frac{R_{1212}}{D}, \qquad R_{1212} = \frac{1}{4}\sum_{p,q} g^{pq}\big(\Phi_{12p}\Phi_{12q} - \Phi_{11p}\Phi_{22q}\big),$$

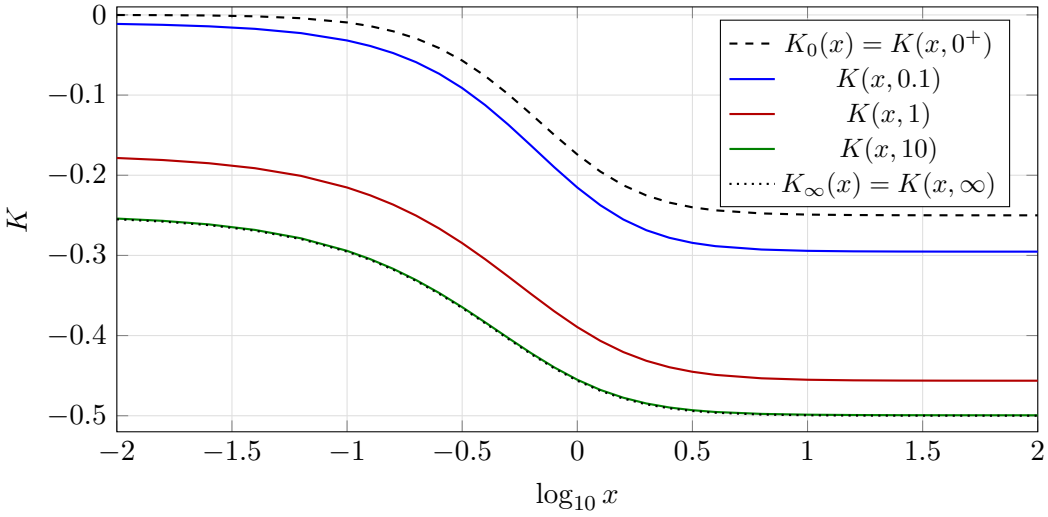



Figure 1: The curvature $K(x,y)$ as a function of $\log_{10} x$ for $y = 0.1, 1, 10$, together with the edge laws $K_0$ and $K_\infty$ of Theorem B. Every curve decreases, and the curves are ordered as $y$ increases. The value $K(x,y) \to K_\infty(y)$ as $x \to \infty$, for example $K(\infty, 0.1) = K_\infty(0.1) = -0.2953\ldots$. Data: `data/plot_K.dat`; illustration only.

*equivalently*

$$K = -\frac{\det M}{4D^2}, \qquad M = \begin{pmatrix} \Phi_{11} & \Phi_{12} & \Phi_{22} \\ \Phi_{111} & \Phi_{112} & \Phi_{122} \\ \Phi_{112} & \Phi_{122} & \Phi_{222} \end{pmatrix}.$$

*Proof.* For a Hessian metric the Christoffel symbols of the first kind are $\Gamma_{ij,k} = \frac{1}{2}\Phi_{ijk}$. In the classical expression

$$R_{1212} = -\tfrac{1}{2}(g_{11,22} - 2g_{12,12} + g_{22,11}) + g^{pq}(\Gamma_{12,p}\Gamma_{12,q} - \Gamma_{11,p}\Gamma_{22,q}), \qquad K = R_{1212}/D,$$

the second-derivative part is $-\frac{1}{2}(\Phi_{1122} - 2\Phi_{1122} + \Phi_{1122}) = 0$. This leaves the stated quadratic expression in third derivatives [9, Ch. 2]. Expanding $g^{pq} = (\operatorname{adj} g)^{pq}/D$ turns $R_{1212}/D$ into $-\det M/(4D^2)$. Both identities were also checked symbolically, together with Brioschi's formula for the Gaussian curvature of $E = \Phi_{11}$, $F = \Phi_{12}$, $G = \Phi_{22}$ (log `01`). □

For the beta manifold $\Phi = \varphi$, we have $\varphi_{11} = \psi_1(x) - \psi_1(s)$, $\varphi_{12} = -\psi_1(s)$ and $\varphi_{22} = \psi_1(y) - \psi_1(s)$, and the third derivatives are obtained by replacing $\psi_1$ by $\psi_2$. Lemma 2.1 then gives (2).

## 2.2 The defect representation

Let $F \in C^4(0,\infty)$ with $F'' > 0 > F'''$ and put

$$\psi_1 := F'', \quad \psi_2 := F''', \quad h := \frac{1}{\psi_1}, \quad G := -\frac{\psi_1}{\psi_2} = \frac{h}{h'}.$$

Then $h' = -\psi_2/\psi_1^2 > 0$. For $x, y > 0$ and $s = x + y$ define the *defects*

$$A(x,y) = G(s) - G(x) - G(y), \qquad B(x,y) = h(s) - h(x) - h(y).$$

**Theorem 2.2** (defect representation)**.** *Let $\Phi(x,y) = F(x) + F(y) - F(x+y)$. The Hessian of $\Phi$ is positive definite at $(x,y)$ if and only if $B(x,y) > 0$, and there*

$$K_F(x,y) = -\frac{h'(x)\,h'(y)\,h'(s)\,A(x,y)}{4\,B(x,y)^2}. \tag{3}$$

*In particular* $\operatorname{sign} K_F = -\operatorname{sign} A$.

*Proof.* Write $p_k = \psi_k(x)$, $q_k = \psi_k(y)$, $r_k = \psi_k(s)$. Then $\Phi_{11} = p_1 - r_1 > 0$, because $\psi_1$ is decreasing. The determinant is

$$D = p_1q_1 - r_1(p_1 + q_1) = p_1q_1r_1\Big(\frac{1}{r_1} - \frac{1}{p_1} - \frac{1}{q_1}\Big) = p_1q_1r_1\,B,$$

which gives the first claim. The matrix $M$ of Lemma 2.1 has rows $(p_1 - r_1, -r_1, q_1 - r_1)$, $(p_2 - r_2, -r_2, -r_2)$, $(-r_2, -r_2, q_2 - r_2)$, and a direct expansion gives

$$-\det M = r_2(p_1q_2 + q_1p_2) - r_1p_2q_2 = p_2q_2r_2\Big(\frac{p_1}{p_2} + \frac{q_1}{q_2} - \frac{r_1}{r_2}\Big) = p_2q_2r_2\,A.$$

Hence $K_F = p_2q_2r_2A/(4p_1^2q_1^2r_1^2B^2)$. Since $p_2/p_1^2 = -h'(x)$, $q_2/q_1^2 = -h'(y)$ and $r_2/r_1^2 = -h'(s)$, this is (3). □

**Corollary 2.3** (negativity)**.** *Suppose $h$ and $G$ extend continuously to $0$ with $h(0) = G(0) = 0$, and $h$, $G$ are strictly convex. Then $A > 0$, $B > 0$ and $K_F < 0$ on $(0,\infty)^2$, and*

$$A = \int_0^x\int_0^y G''(a+b)\,db\,da, \qquad B = \int_0^x\int_0^y h''(a+b)\,db\,da. \tag{4}$$

*Proof.* By the fundamental theorem of calculus, $\int_0^x\int_0^y G''(a+b)\,db\,da = G(s)-G(x)-G(y)+G(0)$. The integrand is positive. □

For the beta manifold the hypotheses hold by Lemma 3.1 and Theorem 3.3 (S1, S3). Corollary 2.3 therefore gives a short proof of the negativity theorem of [4, 5] in dimension two, modulo the shape lemmas.

### 2.3 The logarithmic derivative

Assume $A, B > 0$ and put

$$q(x,y) := \partial_x \log(-K_F(x,y)), \qquad T(z) := \frac{h''(z)}{h'(z)}, \qquad R(z) := T(z) - \frac{1}{z}.$$

Since $-K_F > 0$, we have $\partial_x K_F < 0$ if and only if $q > 0$. We use the increments

$$\Delta G' = G'(s)-G'(x), \qquad \Delta h' = h'(s)-h'(x), \qquad D_\delta = \delta'(x)-\delta'(s), \qquad \delta(z) := G(z)-2h(z)+z.$$

**Proposition 2.4** (raw and balance formulas)**.**

$$q = T(x) + T(s) + \frac{\Delta G'}{A} - \frac{2\Delta h'}{B}, \tag{5}$$

$$q = T(x) + T(s) - \frac{D_\delta}{A} + \frac{2\Delta h'\,(B-A)}{AB}. \tag{6}$$

*Moreover* $2B - A = \delta(x) + \delta(y) - \delta(s)$.

*Proof.* By (3), $\log(-K_F) = \log h'(x)+\log h'(y)+\log h'(s)+\log A-2\log B-\log 4$, and $\partial_x A = \Delta G'$, $\partial_x B = \Delta h'$; this gives (5). Since $G' = 2h' - 1 + \delta'$, we have $\Delta G' = 2\Delta h' - D_\delta$, so

$$\frac{\Delta G'}{A} - \frac{2\Delta h'}{B} = -\frac{D_\delta}{A} + 2\Delta h'\Big(\frac{1}{A} - \frac{1}{B}\Big),$$

which gives (6). Finally $2B-A = -[(G-2h)(s)-(G-2h)(x)-(G-2h)(y)] = \delta(x)+\delta(y)-\delta(s)$, because $z \mapsto z$ is additive. □

*Remark* 2.5 (why "balance"). For $F = \log\Gamma$ we have $\delta(z) = \frac{1}{2} - \frac{11}{360}z^{-3} + O(z^{-4})$ (Lemma 3.2). Hence $2B - A = \frac{1}{2} + O(\min(x,y)^{-3})$, while $A$ and $B$ themselves differ from $\frac{1}{2}$ by terms of order $1/\min(x,y)$. In (5) the two defect terms are each of order $x^{-2}$ and cancel to leading order. In (6) no such cancellation remains: $D_\delta = O(x^{-4})$, and the last term is $O(x^{-3})$ and positive.

**Proposition 2.6** (averaged formula)**.** *Suppose $h, G \in C^3[0,\infty)$ with $h(0) = G(0) = 0$, and $A, B > 0$. Define*

$$\langle f\rangle(x,y) = \int_0^1\!\!\int_0^1 f(\alpha x + \beta y)\, d\beta\, d\alpha, \qquad \langle \alpha f\rangle(x,y) = \int_0^1\!\!\int_0^1 \alpha\, f(\alpha x + \beta y)\, d\beta\, d\alpha.$$

*Then $A = xy\langle G''\rangle$, $B = xy\langle h''\rangle$, and*

$$q = R(x) + T(s) + \frac{\langle \alpha G'''\rangle}{\langle G''\rangle} - \frac{2\langle \alpha h'''\rangle}{\langle h''\rangle}. \tag{7}$$

*Proof.* Substituting $a = \alpha x$ and $b = \beta y$ in (4) gives $A = xy\langle G''\rangle$ and $B = xy\langle h''\rangle$. Hence $\partial_x \log A = \frac{1}{x} + \langle \alpha G'''\rangle/\langle G''\rangle$ (differentiation under the integral sign), and similarly for $B$. Insert these into $q = T(x) + T(s) + \partial_x \log A - 2\partial_x \log B$. □

In (7) the terms $T(x)$, $\partial_x \log A$ and $-2\partial_x \log B$, which are each of size $1/x$ near the edge $x = 0$, have been combined. Every remaining term is regular up to $x = 0$.

# 3 Polygamma toolkit and shape lemmas

From now on $F = \log\Gamma$, so $\psi_k = \psi^{(k)}$ and $h = 1/\psi_1$, $G = -\psi_1/\psi_2$. We use the functions

$$T = \frac{h''}{h'}, \quad R = T - \frac{1}{z}, \quad g = z - G, \quad e = z - h, \quad \hat{g} = \tfrac{1}{2} - g,$$

$$\delta' = G' - 2h' + 1, \quad \varepsilon = 2h'' - G'' = -\delta'', \quad \kappa = G - h.$$

Each of these is a rational function of $z$ and $\psi_1(z), \dots, \psi_7(z)$, obtained by exact symbolic differentiation with $\psi_k' = \psi_{k+1}$ (script `symforms.py`).

## 3.1 Three exact representations

**Lemma 3.1** (regular form near 0)**.** *Put $P_k(z) = \psi_k(1+z)$, which is analytic on $(-1,\infty)$. Then $\psi_k(z) = (-1)^{k+1}k!\, z^{-k-1} + P_k(z)$. In particular*

$$h(z) = \frac{z^2}{1 + z^2 P_1(z)}, \qquad G(z) = \frac{z\,(1 + z^2 P_1(z))}{2 - z^3 P_2(z)},$$

*so $h$ and $G$ are analytic in a neighbourhood of $[0,\infty)$, with $h(0) = h'(0) = 0$, $h''(0) = 2$, $G(0) = 0$, $G'(0) = \frac{1}{2}$, $G''(0) = 0$ and $G'''(0) = 3\zeta(2)$. More generally, every quantity $Q$ above can be written exactly as $Q = z^a\, n(z, P_1, \dots, P_7)/d(z, P_1, \dots, P_7)$, with polynomials $n$, $d$ in $z$ and the $P_k$, $a \in \mathbb{Z}$, and $n(0,\cdot), d(0,\cdot) \not\equiv 0$.*

*Proof.* Use the recurrence $\psi_k(z) = \psi_k(z+1) + (-1)^{k+1}k! z^{-k-1}$, clear denominators, and cancel the largest common power of $z$. The last step is exact polynomial algebra in the independent symbols $z, P_1, \dots, P_7$. □

**Lemma 3.2** (enveloping expansion; Alzer [1])**.** *For integers $k \geq 1$, $n \geq 0$ and real $z > 0$ there is $\theta \in (0,1)$ such that*

$$(-1)^{k+1}\psi_k(z) = S_{k,n}(1/z) + \theta\, E_{k,n}(1/z),$$

*where*

$$S_{k,n}(w) = (k-1)!\,w^k + \tfrac{k!}{2}\,w^{k+1} + \sum_{j=1}^{n} \frac{B_{2j}\,(2j+k-1)!}{(2j)!}\,w^{2j+k},$$

$$E_{k,n}(w) = \frac{B_{2n+2}\,(2n+k+1)!}{(2n+2)!}\,w^{2n+k+2}.$$

*Proof.* From the partial-fraction expansion of coth,

$$\frac{t}{2}\coth\frac{t}{2} = 1 + \sum_{m\geq 1} \frac{2t^2}{t^2+4\pi^2m^2}.$$

With $y_m = t^2/(4\pi^2m^2)$ we have $y_m/(1+y_m) = \sum_{i=1}^{n}(-1)^{i-1}y_m^i + (-1)^n y_m^{n+1}/(1+y_m)$. Summing over $m$ and using $\sum_m 2(2\pi m)^{-2i} = (-1)^{i+1}B_{2i}/(2i)!$ gives, for $t>0$,

$$\frac{t}{2}\coth\frac{t}{2} = \sum_{i=0}^{n} \frac{B_{2i}t^{2i}}{(2i)!} + \vartheta_n(t)\frac{B_{2n+2}t^{2n+2}}{(2n+2)!}, \qquad \vartheta_n(t)\in(0,1).$$

Now $(-1)^{k+1}\psi_k(z) = \int_0^\infty t^{k-1}e^{-zt}\,\frac{t}{1-e^{-t}}\,dt$, and $\frac{t}{1-e^{-t}} = \frac{t}{2} + \frac{t}{2}\coth\frac{t}{2}$. Integrate term by term using $\int_0^\infty t^{k-1+j}e^{-zt}dt = (k-1+j)!\,z^{-k-j}$. The remainder is a positive multiple of $B_{2n+2}$, of absolute value smaller than the corresponding term, because the kernel is positive and $\vartheta_n\in(0,1)$. □

Substituting Lemma 3.2 with $n=4$ into a quantity $Q$ and expanding exactly in $w=1/z$ and the parameters $\theta_1,\dots,\theta_7\in[0,1]$, one for each $\psi_k$, gives $Q = w^m\,\tilde n(w,\theta)/\tilde d(w,\theta)$ with explicit rational polynomials. For every quantity used below, the lowest power of $w$ in $\tilde n$ and in $\tilde d$ occurs only in the $\theta$-free part. The leading behaviour is therefore exact, for instance

$$h'' \sim \tfrac{1}{6}z^{-3}, \quad h''' \sim -\tfrac{1}{2}z^{-4}, \quad G'' \sim \tfrac{1}{3}z^{-3}, \quad G''' \sim -z^{-4},$$

$$\varepsilon \sim \tfrac{11}{30}z^{-5}, \quad \varepsilon' \sim -\tfrac{11}{6}z^{-6}, \quad \kappa' \sim -\tfrac{1}{12}z^{-2}, \quad \delta' \sim \tfrac{11}{120}z^{-4}.$$

These leading terms show that the expansions vanish to high order; in $\varepsilon$ the two leading orders of $2h''$ and $G''$ cancel.

### 3.2 Certified evaluation

A quantity $Q$ is enclosed on an interval $[a,b]$ in one of three ways:

- on $(0,1]$, by the regular form of Lemma 3.1, with $P_k$ enclosed through the Hurwitz zeta function at the ball $1+[a,b]$;
- on $[1,3]$, by the direct rational form, with $\psi_k(z) = (-1)^{k+1}k!\,\zeta(k+1,z)$ enclosed in Arb;
- on $[3,\infty]$, by the enveloping form with $w\in[1/b,1/a]$ and $\theta_k\in[0,1]$.

All arithmetic is midpoint–radius ball arithmetic at 256 bits; a ball comparison is accepted only if it is certain. To certify a sign on an interval, the interval is cut at 1 and 3 and bisected (geometrically on long ranges) until every piece has a certified sign. Since $z^a>0$ and $w^m>0$, the sign of $Q$ on a piece is $\operatorname{sign}(n)\operatorname{sign}(d)$ (respectively $\operatorname{sign}(\tilde n)\operatorname{sign}(\tilde d)$) whenever both are certain. A consistency check of all three forms against high-precision `mpmath` values is in log `02`.

### 3.3 The shape lemmas

**Theorem 3.3** (shape lemmas)**.** *On* $(0,\infty)$*:*

(S1) $h'' > 0$*;*

(S2) $h''' < 0$*;*

(S3) $G'' > 0$*;*

(S4) $G'''' < 0$ *on* $(0, \frac{2}{5}]$*,* $G''' < 0$ *on* $[\frac{3}{10}, \infty)$ *and* $G''' > 0$ *on* $(0, \frac{29}{100}]$*;*

(S5) $\varepsilon = 2h'' - G'' > 0$*;*

(S6) $\varepsilon' = 2h''' - G''' < 0$*;*

(S7) $\kappa' = G' - h' < 0$ *on* $[1,\infty)$ *and* $\kappa > 0$ *on* $(0,1]$*.*

*Proof.* Each statement is a sign condition on an explicit quantity. It was certified by the procedure of the previous subsection (script `lemmas1d.py`, log `03`); the numbers of certified pieces are listed in Table 1. □

| lemma | statement | pieces |
|---|---|---|
| S1 | $h'' > 0$ on $(0,\infty)$ | 301 |
| S2 | $h''' < 0$ on $(0,\infty)$ | 1 020 |
| S3 | $G'' > 0$ on $(0,\infty)$ | 1 562 |
| S4 | $G'''' < 0$ on $(0, \frac{2}{5}]$; $G''' < 0$ on $[\frac{3}{10}, \infty)$; $G''' > 0$ on $(0, \frac{29}{100}]$ | 41; 8 395; 41 |
| S5 | $2h'' - G'' > 0$ on $(0,\infty)$ | 10 835 |
| S6 | $2h''' - G''' < 0$ on $(0,\infty)$ | 34 824 |
| S7 | $G' - h' < 0$ on $[1,\infty)$; $G - h > 0$ on $(0,1]$ | 962; 10 |

Table 1: Certified shape lemmas: number of interval pieces in the final partitions (ball arithmetic, 256 bits; total time 51 s).

**Corollary 3.4.** (a) *$h' > 0$, $h'(0^+) = 0$, $h'(\infty) = 1$, hence $0 < h' < 1$. The function $e = z - h$ is increasing and concave, with $e(0^+) = 0$ and $e(\infty) = \frac{1}{2}$, so $0 < e < \frac{1}{2}$. The function $e' = 1 - h'$ is decreasing. Moreover $h(z) < z^2$.*

(b) *$G'(0^+) = \frac{1}{2}$ and $G'(\infty) = 1$, hence $\frac{1}{2} < G' < 1$. The function $g = z - G$ is increasing and concave, with $g(0^+) = 0$ and $g(\infty) = \frac{1}{2}$, so $0 < g < \frac{1}{2}$. The functions $g'$ and $\hat{g} = \frac{1}{2} - g$ are decreasing.*

(c) *$T$ is strictly decreasing.*

(d) *$\delta'$ is strictly decreasing with $\delta'(\infty) = 0$, so $\delta' > 0$; $\varepsilon$ is positive and strictly decreasing.*

(e) *There is $x_m \in (\frac{29}{100}, \frac{3}{10})$ such that $G''$ is strictly increasing on $(0, x_m]$ and strictly decreasing on $[x_m, \infty)$.*

(f) *$\kappa > 0$ on $(0,\infty)$, $\kappa$ is decreasing on $[1,\infty)$, and $\kappa(\infty) = 0$.*

(g) *For fixed $y > 0$, $x \mapsto \Delta G'(x,y) = G'(x+y) - G'(x)$ is first increasing and then decreasing (possibly only one of the two). In particular, on every interval its minimum is attained at an endpoint.*

*Proof.* The limits at 0 follow from the regular forms of Lemma 3.1. The limits at $\infty$ follow from Lemma 3.2 applied to each $\psi_k$ separately, inserted into the exact rational expressions for $h, h', G, G', \dots$; no asymptotic series has to be differentiated term by term. This gives, for example, $h = z - \frac{1}{2} + \frac{1}{12z} + O(z^{-2})$, $h' = 1 + O(z^{-2})$, $G = z - \frac{1}{2} + \frac{1}{6z} + O(z^{-2})$ and $G' = 1 + O(z^{-2})$.

(a) $h' = -\psi_2/\psi_1^2 > 0$. By S1, $h'$ increases from 0 to 1, and $e'' = -h'' < 0$. The bound $h(z) < z^2$ holds because $\psi_1(z) > z^{-2}$.

(b) follows in the same way from S3.

(c) $T' = (h'''h' - h''^2)/h'^2 < 0$ by S1 and S2.

(d) $\delta'' = -\varepsilon < 0$ by S5, $\delta'(\infty) = 1 - 2 + 1 = 0$, and $\varepsilon' < 0$ by S6.

(e) By S4 the function $G'''$ is strictly decreasing on $(0, \frac{2}{5}]$, positive at $\frac{29}{100}$ and negative on $[\frac{3}{10}, \infty)$. So it has exactly one zero $x_m$, and $x_m \in (\frac{29}{100}, \frac{3}{10})$.

(f) $\kappa$ decreases on $[1, \infty)$ to $\kappa(\infty) = 0$, so $\kappa > 0$ there; on $(0, 1]$ this is S7.

(g) We have $\partial_x \Delta G' = G''(x + y) - G''(x)$. By (e), this is positive if $x + y \le x_m$, negative if $x \ge x_m$, and strictly decreasing in $x$ on $(x_m - y, x_m)$, where $G''(x + y)$ decreases and $G''(x)$ increases. So it changes sign at most once, from + to −. □

# 4 Positivity of the logarithmic derivative

## 4.1 Regions

Fix the parameters

$$\sigma_c = \tfrac{7}{25}, \qquad \varepsilon_L = \tfrac{1}{10}, \qquad \varepsilon_B = \tfrac{1}{50}, \qquad L = 4, \qquad V = 10,$$

and define the regions

$$\begin{aligned}
\mathsf{C} &= \{x + y \le \sigma_c\} && \text{(corner)},\\
\mathsf{Lf} &= \{x \le \varepsilon_L,\ y \ge \sigma_c - \varepsilon_L\} && \text{(left strip)},\\
\mathsf{Bt} &= \{y \le \varepsilon_B,\ \sigma_c - \varepsilon_B \le x \le L\} && \text{(bottom strip)},\\
\mathsf{I} &= \{x \ge L\} && \text{(large } x\text{)},\\
\mathsf{II} &= \{\varepsilon_L \le x \le L,\ y \ge V\} && \text{(large } y\text{)},\\
\mathsf{M} &= [\varepsilon_L, L] \times [\varepsilon_B, V] \cap \{x + y \ge \sigma_c\} && \text{(main region)}.
\end{aligned}$$

**Lemma 4.1** (coverage)**.** $(0, \infty)^2 = \mathsf{C} \cup \mathsf{Lf} \cup \mathsf{Bt} \cup \mathsf{I} \cup \mathsf{II} \cup \mathsf{M}$.

*Proof.* Let $x, y > 0$. If $x \ge L$, the point is in $\mathsf{I}$. If $x < \varepsilon_L$, it is in $\mathsf{Lf}$ when $y \ge \sigma_c - \varepsilon_L$, and otherwise $x + y < \sigma_c$. Now let $\varepsilon_L \le x < L$. If $y \ge V$, the point is in $\mathsf{II}$. If $y < \varepsilon_B$, it is in $\mathsf{Bt}$ when $x \ge \sigma_c - \varepsilon_B$, and otherwise $x + y < \sigma_c$. If $\varepsilon_B \le y < V$, it is in $\mathsf{M}$ when $x + y \ge \sigma_c$, and in $\mathsf{C}$ otherwise. □

In each region we bound $q$ from below by an expression that is either a function of one variable or, in $\mathsf{M}$, a function of the corners of a box, and we certify that this bound is positive. Throughout, "decreasing" and the like refer to Corollary 3.4.

## 4.2 The corner

**Lemma 4.2.** *On* $\mathsf{C}$, $q(x, y) \ge \frac{1}{s} + R(x) + R(s)$, *and this is* $> 0$.

*Proof.* Use (7). The points $\alpha x + \beta y$ lie in $(0, s] \subset (0, \frac{29}{100}]$, where $G''' > 0$ (S4), so $\langle \alpha G''' \rangle > 0$. Also $h''' < 0$ (S2), so $-2\langle \alpha h''' \rangle / \langle h'' \rangle > 0$. Finally $T(s) = \frac{1}{s} + R(s)$. For positivity, split $(0, \sigma_c]$ into 400 equal pieces $(s_{k-1}, s_k]$. Let $m_k$ be a certified lower bound of $R$ on $(0, s_k]$, obtained from the regular form of Lemma 3.1 ($R = z \cdot n/d$). For $s \in (s_{k-1}, s_k]$ we then have $q \ge \frac{1}{s_k} + 2m_k$, and the certified minimum of these 400 numbers is 1.7808 (log 04). □

### 4.3 The left strip

**Lemma 4.3.** *Let $R_* = \min_{[0,\varepsilon_L]} R$ and $\mu(y) = \min_{[y,y+\varepsilon_L]} G''$. On* Lf*,*

$$q(x,y) \;\geq\; R_* + T(y+\varepsilon_L) + \min\Big\{0, \frac{\mu(y) - G''(\varepsilon_L)}{2\,(G'(y) - G'(\varepsilon_L))}\Big\} + \frac{h''(\varepsilon_L) - h''(y)}{h'(y)},$$

*and the right-hand side is positive for all $y \geq \sigma_c - \varepsilon_L = \frac{9}{50}$.*

*Proof.* Use (7) and bound its four terms in turn. First, $R(x) \geq R_*$. Second, $T(s) \geq T(y+\varepsilon_L)$, because $T$ is decreasing. Third, integrating in $\beta$,

$$\langle \alpha G''' \rangle = \int_0^1 \alpha\, \frac{G''(\alpha x + y) - G''(\alpha x)}{y}\, d\alpha \;\geq\; \frac{\mu(y) - G''(\varepsilon_L)}{2y},$$

because $G''$ is increasing on $[0,\varepsilon_L] \subset [0,x_m]$. Likewise $\langle G'' \rangle = \int_0^1 (G'(\alpha x + y) - G'(\alpha x))\, d\alpha / y \geq (G'(y) - G'(\varepsilon_L))/y > 0$. A lower bound $N \leq \langle \alpha G''' \rangle$ with $N < 0$, together with $\langle G'' \rangle \geq D_0 > 0$, gives $\langle \alpha G''' \rangle / \langle G'' \rangle \geq \min\{0, N/D_0\}$. Fourth, since $h''$ is decreasing,

$$-\langle \alpha h''' \rangle = \int_0^1 \alpha\, \frac{h''(\alpha x) - h''(\alpha x + y)}{y}\, d\alpha \geq \frac{h''(\varepsilon_L) - h''(y)}{2y} > 0.$$

Also $\langle h'' \rangle \leq h'(y)/y$, since $h'(c+y) - h'(c) = \int_c^{c+y} h'' \leq \int_0^y h'' = h'(y)$ for $c \geq 0$. This proves the inequality.

For positivity, $[\frac{9}{50}, 10^6]$ is partitioned geometrically with ratio 1.02 into 785 intervals $[y_0, y_1]$. On each interval we use $T(y+\varepsilon_L) \geq T(y_1+\varepsilon_L)$, $G'(y) \geq G'(y_0)$, $h''(y) \leq h''(y_0)$, $h'(y) \leq h'(y_1)$, and $\mu \geq \min\{G''(y_0), G''(y_1+\varepsilon_L)\}$, the last by unimodality. For $y \geq 10^6$ we use $T \geq 0$, $\mu \geq 0$ and $h' < 1$. The certified minimum of the bound is 1.0038 (log `04`). □

### 4.4 The bottom strip

**Lemma 4.4.** *Let $\sigma_c - \varepsilon_B \leq x_0 \leq x \leq x_1 \leq L$, $0 < y \leq \varepsilon_B$, and $h'(x_0) > \varepsilon_B$. Then*

$$q(x,y) \;\geq\; T(x_1) + T(x_1+\varepsilon_B) + \frac{\min_{[x_0, x_1+\varepsilon_B]} G''}{G'(x_1+\varepsilon_B) - \frac{1}{2}} - \frac{2h''(x_0)}{h'(x_0) - \varepsilon_B},$$

*and the right-hand side is positive on a partition of $[\frac{13}{50}, 4]$.*

*Proof.* Use (5) and write the increments and defects as integrals in the small variable:

$$\Delta G' = y\int_0^1 G''(x+\beta y)d\beta, \qquad A = \int_0^y (G'(x+b) - G'(b))db = y\int_0^1 (G'(x+\beta y) - G'(\beta y))d\beta,$$

and similarly for $\Delta h'$ and $B$. Since $G'$ is increasing and $G'(0^+) = \frac{1}{2}$, the second integral is at most $G'(x+y) - \frac{1}{2}$. Since $h'(x+\beta y) \geq h'(x)$ and $\int_0^1 h'(\beta y)d\beta = h(y)/y < y$, the $B$-integral is at least $h'(x) - y$. Hence

$$\frac{\Delta G'}{A} \geq \frac{\min_{[x,x+y]} G''}{G'(x+y) - \frac{1}{2}}, \qquad \frac{\Delta h'}{B} \leq \frac{h''(x)}{h'(x) - y},$$

and the claim follows from the monotonicity of $T$, $G'$, $h''$ and $h'$. The minimum of $G''$ on an interval is attained at an endpoint by unimodality. With 275 geometric intervals (ratio 1.01), the certified minimum of the bound divided by $T(x_1)$ is 1.4405 (log `04`). □

### 4.5 Large $x$

**Lemma 4.5.** *For $x \geq L$ and every $y > 0$,*

$$q(x,y) \;\geq\; T(x) - \max\Big\{\frac{\varepsilon(x)}{g(1)-g'(x)},\; \frac{\delta'(x)}{g(1)-\hat{g}(x)}\Big\},$$

*provided both denominators are positive. The right-hand side is positive for all $x \geq 4$.*

*Proof.* Use (6). Here $T(s) > 0$ and $\Delta h' > 0$. Also $B - A = \kappa(x) + \kappa(y) - \kappa(s) \geq \kappa(y) > 0$, since $\kappa$ is decreasing on $[1,\infty) \ni x$ and $\kappa > 0$. Hence $q \geq T(x) - D_\delta/A$. Now $D_\delta = \int_x^s \varepsilon$ and $A = g(y) - (g(s) - g(x))$, and there are two cases.

- If $y \leq 1$: since $\varepsilon$ is decreasing, $D_\delta \leq y\,\varepsilon(x)$. By concavity with $g(0) = 0$, $g(y) \geq y\,g(1)$, and $g(s) - g(x) \leq y\,g'(x)$. Hence $D_\delta/A \leq \varepsilon(x)/(g(1) - g'(x))$.
- If $y \geq 1$: since $\delta' > 0$, $D_\delta \leq \delta'(x)$. Also $g(y) \geq g(1)$ and $g(s) - g(x) \leq \frac{1}{2} - g(x) = \hat{g}(x)$. Hence $D_\delta/A \leq \delta'(x)/(g(1) - \hat{g}(x))$.

Here $g(1) = 1 - \zeta(2)/(2\zeta(3)) = 0.3157\ldots$. On $[4, 64]$ we use 141 geometric intervals and endpoint values, where $T$, $\varepsilon$, $g'$, $\delta'$ and $\hat{g}$ are all decreasing. On $[64,\infty)$ we use the enveloping forms on 64 intervals of $w \in [0, \frac{1}{64}]$, after division by $w^3$. The certified relative margins are at least 0.40 and 0.97 (log `04`). □

### 4.6 Large $y$

**Lemma 4.6.** *For $\varepsilon_{\mathrm{L}} \leq x \leq L$ and $y \geq V$,*

$$q(x,y) \;\geq\; T(x) - \frac{\delta'(x)}{g(x) - x\,g'(V)} + \frac{2\,(e'(x) - e'(V))\,\kappa(x)}{g(x)\,e(x)},$$

*provided $g(x) - xg'(V) > 0$. The right-hand side is positive on $[\frac{1}{10}, 4]$.*

*Proof.* Use (6). Here $T(s) > 0$, $D_\delta \leq \delta'(x)$ and

$$A = g(x) - (g(s) - g(y)) \in \big[g(x) - x\,g'(y),\; g(x)\big],$$

with $g'(y) \leq g'(V)$. Also $B \leq e(x)$. Next, $\Delta h' = e'(x) - e'(s) \geq e'(x) - e'(V) > 0$ and $B - A = \kappa(x) + (\kappa(y) - \kappa(s)) \geq \kappa(x) > 0$, because $\kappa$ is decreasing on $[1,\infty) \ni y$. With 371 geometric intervals (ratio 1.01) and endpoint values, the certified minimum of the bound divided by $T(x_1)$ is 0.0884 (log `04`). □

### 4.7 The main region

**Lemma 4.7** (corner bound)**.** *For $0 < x_0 < x_1$, $0 < y_0 < y_1$ and all $(x,y) \in [x_0,x_1] \times [y_0,y_1]$,*

$$q(x,y) \;\geq\; T(x_1) + T(x_1+y_1) + \frac{\min\{\Delta G'(x_0,y_0),\; \Delta G'(x_1,y_0)\}}{A(x_1,y_1)} - \frac{2\,\Delta h'(x_0,y_1)}{B(x_0,y_0)}.$$

*Proof.* Use (5). The function $T$ is decreasing. The defects $A$ and $B$ increase in both variables, because $\partial_x A = G'(s) - G'(x) > 0$, $\partial_y A = G'(s) - G'(y) > 0$, and likewise for $B$. The increment $\Delta G'$ is increasing in $y$, and unimodal in $x$ by Corollary 3.4(g). The increment $\Delta h'$ is increasing in $y$ and decreasing in $x$, because $h''$ is positive and decreasing. All four terms are therefore bounded by their values at the indicated corners. □

**Proposition 4.8.** *$q > 0$ on* M.

*Proof.* Take rational nodes $x_i$ (respectively $y_j$) forming a geometric progression of ratio about $e^{0.04}$ from $\frac{1}{10}$ to 4 (respectively from $\frac{1}{50}$ to 10). Every box with $x_{i+1} + y_{j+1} \geq \sigma_c$ is certified. Boxes with $x_{i+1} + y_{j+1} < \sigma_c$ lie inside C. The node values of $G, G', h, h', T$ are computed in ball arithmetic at 128 bits. For each of the 13 573 boxes, the right-hand side of Lemma 4.7 is certified positive without any further subdivision. The smallest certified lower bound, relative to $T(x_1)$, is 0.0598, attained near $(x, y) = (0.10, 1.75)$ (log `05`). An independent run with ratio $e^{0.02}$ (53 747 boxes) also succeeds, with smallest relative bound 0.105 (log `06`). As a second, formula-independent certificate, the analogous corner bound obtained from the balance formula (6) also certifies all of M: 27 489 boxes after 4 678 adaptive bisections (log `08`). This bound uses $D_\delta$, which is increasing in $y$ and decreasing in $x$ by S5–S6, and the sign of $B - A$ on the box. □

### 4.8 Conclusion

**Theorem 4.9.** *$q(x, y) = \partial_x \log(-K(x, y)) > 0$ for all $x, y > 0$.*

*Proof.* Combine Lemmas 4.1, 4.2, 4.3, 4.4, 4.5 and 4.6 with Proposition 4.8. Table 2 summarizes the certificates. □

| region | formula | partition | certified minimum |
|---|---|---|---|
| C: $x + y \leq \frac{7}{25}$ | (7) | 400 pieces of $(0, \frac{7}{25}]$ | $\frac{1}{s} + 2 \min R \geq 1.780$ |
| Lf: $x \leq \frac{1}{10}$ | (7) | 785 intervals in $y$, plus tail | bound $\geq 1.003$ |
| Bt: $y \leq \frac{1}{50}$ | (5) | 275 intervals in $x$ | bound$/T(x_1) \geq 1.440$ |
| I: $x \geq 4$ | (6) | 141 intervals, plus 64 in $w = 1/x$ | relative margin $\geq 0.40$ |
| II: $y \geq 10$ | (6) | 371 intervals in $x$ | bound$/T(x_1) \geq 0.088$ |
| M | (5) | 13 573 boxes (53 747 recheck) | bound$/T(x_1) \geq 0.059$ |
| M (recheck) | (6) | 27 489 boxes | bound$/T(x_1) \geq 2.4 \cdot 10^{-5}$ |

Table 2: Certificates for $q > 0$ (logs `04`–`06`, `08`). All bounds are rigorous ball-arithmetic lower bounds; the minima are rounded down.

## 5 Proofs of Theorems A and B and Corollary D

*Proof of Theorem A.* By Corollary 2.3 we have $K < 0$. By Theorem 4.9, $\partial_x(-K) = q \cdot (-K) > 0$, so $\partial_x K < 0$, and by the symmetry $K(x, y) = K(y, x)$ also $\partial_y K < 0$.

As $\min(x, y) \to \infty$ we have $h'(x), h'(y), h'(s) \to 1$ and $A = g(x) + g(y) - g(s) \to \frac{1}{2}$, and likewise $B \to \frac{1}{2}$. So $K \to -\frac{1/2}{4 \cdot 1/4} = -\frac{1}{2}$ by (3). Because $K$ decreases strictly in each variable,

$$K(x, y) > K(x + 1, y + 1) \geq \lim_{N \to \infty} K(x + N, y + N) = -\tfrac{1}{2}.$$

Near 0, Lemma 3.1 gives $h'(z) = 2z + O(z^3)$, $B = 2xy(1 + O(s^2))$ and $A = \frac{3}{2}\zeta(2)\, xys\,(1 + O(s))$. Hence $K(x, y) = -\frac{3}{4}\zeta(2)s^2(1 + O(s)) \to 0$ as $s \to 0$. Thus $\sup K = 0$ and $\inf K = -\frac{1}{2}$, and neither is attained. □

*Proof of Theorem B. Limit $y \to 0^+$.* As $y \to 0^+$ we have $h'(y)/y \to h''(0) = 2$, $A/y \to G'(x) - G'(0^+) = G'(x) - \frac{1}{2}$ and $B/y \to h'(x)$. Writing (3) as

$$-K = \frac{h'(x)h'(s)\,(h'(y)/y)\,(A/y)}{4(B/y)^2},$$

we get $K \to -\frac{1}{2}(G'(x) - \frac{1}{2})$.

*Limit $y \to \infty$.* As $y \to \infty$ we have $h'(y), h'(s) \to 1$, $A = g(x) - (g(s) - g(y)) \to g(x)$ and $B \to e(x)$, which gives $K_\infty$.

*Monotonicity and ranges.* $K_0$ is strictly decreasing with range $(-\frac{1}{4}, 0)$, because $G'' > 0$ and $G'$ maps $(0, \infty)$ onto $(\frac{1}{2}, 1)$. By Theorem A, $K(x, \cdot)$ is strictly decreasing, so $K_\infty(x) < K(x, y) < K_0(x)$, and by symmetry the same holds with $x$ and $y$ interchanged. For the strict decrease of $K_\infty$, observe that $\log(-K_\infty(x)) = \lim_{y\to\infty} \log(-K(x, y))$ and that $q(x, y)$ is bounded below by a positive function of $x$ alone, uniformly for large $y$:

- by Lemma 4.3 for $x \le \varepsilon_{\mathrm{L}}$;
- by Lemma 4.6 for $\varepsilon_{\mathrm{L}} \le x \le L$;
- by Lemma 4.5 for $x \ge L$.

Integrating in $x$ gives the strict decrease. Finally, as $x \to 0^+$ we have $h'(x) \sim 2x$, $g(x) \sim x/2$ and $e(x) \sim x$, so $K_\infty(0^+) = -\frac{1}{4}$; and $K_\infty(\infty) = -\frac{1}{2}$ because $h' \to 1$ and $g, e \to \frac{1}{2}$. □

*Proof of Corollary D.* We have $\frac{d}{dx}K(x, x) = (\partial_x K + \partial_y K)(x, x) < 0$ and $\frac{d}{d\lambda}K(\lambda x, \lambda y) = x\,\partial_x K + y\,\partial_y K < 0$. The limits 0 and $-\frac{1}{2}$ come from Theorem A. Level sets are graphs by the implicit function theorem, since $\partial_y K \neq 0$, and their slope $-\partial_x K/\partial_y K$ is negative. □

## 6 Remarks

*Remark* 6.1 (asymptotics of $q$). A formal expansion using Lemma 3.2 and (6) gives, with $a = 1/x$, $b = 1/y$, $c = 1/s$ and all three small,

$$q(x, y) \approx \tfrac{1}{6}(a^3 + c^3) + \tfrac{1}{18}(a^2 - c^2)(a + b - c) \quad \text{(up to terms of order four)}.$$

In the raw formula (5) the two defect terms are $\pm\frac{1}{3}(a^2 - c^2) + O(3)$ and cancel to leading order. Near the origin $q = \frac{2}{s}(1 + O(s))$. Along the edge $x \to 0^+$, formula (7) gives

$$q(0^+, y) = \frac{2}{h'(y)} + \frac{G''(y)}{2G'(y) - 1} \ (> 2),$$

and along $y \to 0^+$, $q(x, 0^+) = G''(x)/(G'(x) - \frac{1}{2})$. These expansions are confirmed numerically in log `07`; they are not used in the proofs.

*Remark* 6.2 (what is new). To our knowledge, the defect representation (3), the edge laws of Theorem B, the balance identity $2B - A = \delta(x) + \delta(y) - \delta(s)$ and the averaged formula (7) have not appeared before. They hold for any $F$ with $F'' > 0 > F'''$ (the averaged formula under the stated condition at 0). For $F = \log\Gamma$ they turn a two-variable sign problem with three interacting polygamma arguments into shape statements about two one-variable functions, $h = 1/\psi_1$ and $G = -\psi_1/\psi_2$. Earlier approaches to the diagonal [6, 7, 8] relied on complete monotonicity in one variable. They do not extend directly, because $K$ is not a function of one variable and the cancellations differ in different corners of the quadrant: of order $1/x$ along the edges, second order at infinity, and of order $1/s$ at the origin.

*Remark* 6.3 (scope and open questions). The same calculus applies to the Hessian metric of any additive difference $F(x) + F(y) - F(x + y)$: the edge laws, Problem 1.2, and the shape-constrained certification only require shape lemmas for $1/F''$ and $-F''/F'''$. We did not attempt the higher-dimensional Dirichlet manifolds of [5]. There the sectional curvatures of coordinate planes involve more than one defect, and their extremal values and monotonicity are open as far as we know. A conceptual (computer-free) proof of Theorem 4.9 would also be desirable. The margins in Section 4 suggest that the corner bound of Lemma 4.7 with a few more one-variable lemmas might suffice.

# A Reproducibility

All files are in the folder `xxv/` of the accompanying repository. An independent audit re-derived every identity and inequality and stress-tested all region bounds against exact values of $q$ at random points and on random boxes (no violations). It also re-ran the certificates from a separate copy, with identical logs, and checked the Arb semantics used. It found no error affecting the proof; its minor suggestions (cache versioning, clearer diagnostics) are incorporated. Software: Python 3.13, SymPy, mpmath, and python-flint 0.9 (Arb ball arithmetic [3]). The command `sh scripts/run_all.sh` reproduces every certificate and writes the logs in about two minutes on one core; the first run also builds the exact symbolic forms (`data/forms.pkl`).

| | |
|---|---|
| `symforms.py` | exact symbolic forms: direct, regular near 0 (Lemma 3.1), enveloping (Lemma 3.2, $n = 4$) |
| `rig1d.py`, `cert1d.py` | ball evaluation of the forms; sign certification by interval splitting |
| `lemmas1d.py` | Theorem 3.3 (log `03`) |
| `regions1d.py` | Lemmas 4.2–4.6 (log `04`) |
| `mainregion.py` | Proposition 4.8 (logs `05`, `06`, `08`) |
| `curvsym.py` | Lemma 2.1, formula (2), Theorem 2.2 (log `01`) |
| `sanity.py` | cross-check of all forms against `mpmath` (log `02`) |
| `numerics.py` | values, edge laws, asymptotics and figure data (log `07`; illustration only) |

# B Sample values

| $(x, y)$ | $K(x, y)$ | remark |
|---|---|---|
| $(1, 1)$ | $-0.389403045225$ | |
| $(0.5, 2)$ | $-0.376262957898$ | |
| $(10, 10)$ | $-0.498677032543$ | |
| $(0.01, 0.01)$ | $-0.000469323266$ | $\approx -\frac{3}{4}\zeta(2)(x+y)^2$ |
| $(50, 0.3)$ | $-0.361459896167$ | $\approx K_\infty(0.3)$ |
| $(1, 1000)$ | $-0.456303568529$ | $\approx K_\infty(1) = -0.456303691\ldots$ |

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